12

Inner Products and Orthogonality

Historical Note

The infinite-dimensional case came first. Fourier, in work of 1807 that his examiners refused to publish and that finally appeared in 1822, expanded arbitrary functions in sines and cosines, and the coefficients he wrote down are inner products with an orthogonal family, though nobody would have described them that way for another century. The inequality of Section 12.2 was proved for sums by Cauchy in 1821, for integrals by Bunyakovsky in 1859, and again by Schwarz in 1888, which is why the Russian literature calls it Cauchy–Bunyakovsky–Schwarz and everyone else drops a name. The orthogonalisation procedure of Section 12.4 is usually credited to Gram in 1883 and Schmidt in 1907, and it appears in Laplace's book on probability of 1812 and in Cauchy's work of 1836; the attribution is simply wrong, and universal. Only with Hilbert's study of integral equations after 1904, and von Neumann's axioms for quantum mechanics in 1929, did the abstract notion of an inner product space arrive to explain what all these computations had in common.

Chapter 2 measured length and angle in $\mathbb{R}^n$ using the dot product, and nothing since Chapter 5 has needed them: spans, bases, rank, determinants and eigenvalues are all defined without any notion of length. This chapter puts the geometry back, abstractly, and the payoff is immediate. Once a basis can be required to be orthonormal, coordinates stop being a linear system to solve and become a list of inner products, and changes of basis stop costing accuracy.

12.1 Inner products

Definition 12.1 (Real inner product space)**.** *An* inner product *on a real vector space V is a function $\langle \cdot, \cdot \rangle : V \times V \to \mathbb{R}$ such that for all $u, u', v \in V$ and $\lambda, \mu \in \mathbb{R}$:*

(P1) $\langle u, v \rangle = \langle v, u \rangle$;

(P2) $\langle \lambda u + \mu u', v \rangle = \lambda \langle u, v \rangle + \mu \langle u', v \rangle$;

(P3) $\langle v, v\rangle \geq 0$, *with* $\langle v, v\rangle = 0$ *only for* $v = \mathbf{0}$.

A vector space carrying one is an inner product space, *and we write* $\|v\| = \sqrt{\langle v, v\rangle}$ *for the associated* norm.

Linearity in the second argument follows from (P1) and (P2), so the form is linear in each slot separately; a function of two arguments with that property is called *bilinear*. Property (P3) is the one with content and is called *positive definiteness*.

Example 12.2 (Inner products in the wild). *(i) The dot product* $\langle x, y\rangle = \sum_i x_i y_i$ *on* $\mathbb{R}^n$, *from Definition 2.8. This is the prototype and the default.*

(ii) A weighted *product* $\langle x, y\rangle_w = \sum_i w_i x_i y_i$ *with all* $w_i > 0$. *Weighting coordinates differently is what one does when they are measured in different units, and the positivity of the weights is exactly (P3).*

(iii) On $M_{m\times n}(\mathbb{R})$, *the* Frobenius *product* $\langle A, B\rangle = \operatorname{trace}(A^t B) = \sum_{i,j} a_{ij} b_{ij}$, *which is the dot product of the two matrices flattened into long vectors.*

(iv) On $C[0,1]$, $\langle f, g\rangle = \int_0^1 f(t)g(t)\,dt$. *Positive definiteness needs continuity: a continuous nonnegative function with integral zero is identically zero.*

(v) On P_n, *fix* $n+1$ *distinct points* $t_0, \ldots, t_n$ *and set* $\langle p, q\rangle = \sum_k p(t_k)q(t_k)$. *This is positive definite because a nonzero polynomial of degree at most* n *cannot vanish at* $n+1$ *points.*

Caution

Drop positive definiteness and the geometry collapses. On $\mathbb{R}^2$ the bilinear symmetric form $\langle x, y\rangle = x_1y_1 - x_2y_2$ satisfies (P1) and (P2) but not (P3): the nonzero vector $(1,1)$ has $\langle v, v\rangle = 0$, so it would have length zero. Such indefinite forms are not a mistake, they are the geometry of special relativity, where those null vectors are the paths of light. But nothing in this chapter applies to them, and the classification of what can happen instead is the theory of signature, which is Chapter 15.

12.2 Length, angle and the Cauchy–Schwarz inequality

In Chapter 2 we had the plane in front of us, so we could define the angle geometrically and *derive* the inequality from $|\cos\theta| \leq 1$. Abstractly there is no picture and the order must be reversed: prove the inequality first, then use it to define the angle.

Theorem 12.3 (Cauchy–Schwarz). *For all u, v in an inner product space,*

$$|\langle u, v\rangle| \le \|u\|\,\|v\|,$$

with equality if and only if u and v are linearly dependent.

Proof. If $v = \mathbf{0}$ both sides vanish and u, v are dependent, so assume $v \neq \mathbf{0}$. For every $t \in \mathbb{R}$, positive definiteness gives

$$0 \;\le\; \|u - tv\|^2 \;=\; \|u\|^2 - 2t\,\langle u, v\rangle + t^2\,\|v\|^2,$$

expanding the inner product of $u - tv$ with itself. This holds for all t; choose the t that makes the right side smallest, namely $t = \langle u, v\rangle / \|v\|^2$, which is legitimate because $\|v\| \neq 0$. Substituting,

$$0 \;\le\; \|u\|^2 - \frac{\langle u, v\rangle^2}{\|v\|^2},$$

and multiplying by $\|v\|^2$ gives $\langle u, v\rangle^2 \le \|u\|^2\,\|v\|^2$. Equality forces $\|u - tv\| = 0$ for that t, hence $u = tv$ by (P3); conversely if $u = tv$ then both sides equal $|t|\,\|v\|^2$. □

Corollary 12.4 (Norm properties). $\|\lambda v\| = |\lambda|\,\|v\|$, $\|v\| = 0$ *only for* $v = \mathbf{0}$, *and*

$$\|u + v\| \le \|u\| + \|v\|.$$

Proof. The first two are immediate from (P2) and (P3). For the third, $\|u + v\|^2 = \|u\|^2 + 2\,\langle u, v\rangle + \|v\|^2 \le \|u\|^2 + 2\,\|u\|\,\|v\| + \|v\|^2 = (\|u\| + \|v\|)^2$ by Theorem 12.3. □

Definition 12.5 (Angle and orthogonality). *For nonzero u, v the* angle *between them is the unique $\theta \in [0, \pi]$ with*

$$\cos\theta = \frac{\langle u, v\rangle}{\|u\|\,\|v\|},$$

which makes sense because Theorem 12.3 puts the right side in $[-1, 1]$. We call u and v orthogonal, *written $u \perp v$, when $\langle u, v\rangle = 0$.*

Example 12.6 (An angle between polynomials). *In $C[0,1]$ with $\langle f, g\rangle = \int_0^1 f\,g$, take $f(t) = 1$ and $g(t) = t$. Then $\langle f, g\rangle = \tfrac{1}{2}$, $\|f\| = 1$ and $\|g\| = \sqrt{\int_0^1 t^2} = 1/\sqrt{3}$, so $\cos\theta = \tfrac{1}{2}\sqrt{3} \approx 0.866$ and $\theta = \pi/6$. The constant function and the identity function are*

thirty degrees apart. Nothing in the definition required the vectors to be arrows, and the number is as meaningful here as it is in the plane: it says that t is mostly constant, in the sense that most of it is explained by its average.

12.3 Orthogonal and orthonormal sets

Proposition 12.7 (Orthogonality implies independence)**.** *If $v_1, \dots, v_k$ are nonzero and pairwise orthogonal, they are linearly independent. Moreover Pythagoras holds: $\|v_1 + \cdots + v_k\|^2 = \|v_1\|^2 + \cdots + \|v_k\|^2$.*

Proof. Suppose $\sum_i c_i v_i = \mathbf{0}$ and take the inner product with v_j. All terms with $i \neq j$ vanish by orthogonality, leaving $c_j \|v_j\|^2 = 0$, and $\|v_j\| \neq 0$ forces $c_j = 0$. For the second statement, expand $\langle \sum_i v_i, \sum_j v_j \rangle = \sum_{i,j} \langle v_i, v_j \rangle$ and discard the off-diagonal terms. □

Definition 12.8 (Orthonormal)**.** *A set is* orthonormal *if its members are pairwise orthogonal and each has norm 1; that is, $\langle e_i, e_j \rangle = \delta_{ij}$. An* orthonormal basis *is an orthonormal set that is a basis, which by Proposition 12.7 and Theorem 6.14(c) means simply an orthonormal set of n vectors in an n-dimensional space.*

Theorem 12.9 (Coordinates in an orthonormal basis)**.** *Let $(e_1, \dots, e_n)$ be an orthonormal basis of V. Then for every $v \in V$,*

$$v = \langle v, e_1 \rangle e_1 + \cdots + \langle v, e_n \rangle e_n, \qquad \|v\|^2 = \sum_{i=1}^{n} \langle v, e_i \rangle^2,$$

and for any u, v, $\langle u, v \rangle = \sum_i \langle u, e_i \rangle \langle v, e_i \rangle$.

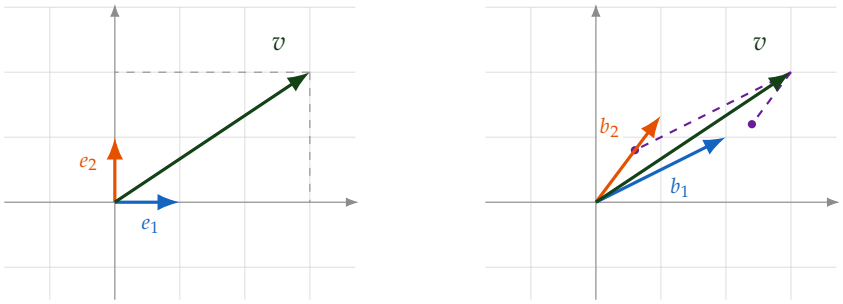
Proof. Write $v = \sum_i c_i e_i$, which is possible and unique because the e_i form a basis. Taking the inner product with e_j gives $\langle v, e_j \rangle = \sum_i c_i \langle e_i, e_j \rangle = c_j$. The other two identities follow by expanding $\langle v, v \rangle$ and $\langle u, v \rangle$ in the same way and using $\langle e_i, e_j \rangle = \delta_{ij}$. □

Key Idea

Compare this with Example 6.21, where finding coordinates in the basis $\big((1,1),(1,-1)\big)$ meant solving a system of two equations. In an orthonormal

basis there is no system: each coordinate is a single inner product, computed independently of all the others, and the norm of the vector is recovered from them by Pythagoras. That is the whole practical argument for orthonormality, and it is why the rest of this part is largely about how to obtain one.

Figure 12.1 puts the two situations side by side. In the orthonormal basis a coordinate is read off by dropping a perpendicular; in the skew basis it is found by running a line through the vector parallel to the other basis direction, which is what solving the system amounts to.



orthonormal
$v = \langle v, e_1\rangle\, e_1 + \langle v, e_2\rangle\, e_2$

skew
solve $c_1 b_1 + c_2 b_2 = v$

Figure 12.1. Theorem 12.9 against Example 6.21. On the left the basis is orthonormal, so dropping perpendiculars reads the coordinates straight off, one inner product each. On the right the basis is skew, so a coordinate is found by running a line through v parallel to one basis vector and seeing where it cuts the other axis. Those two marked intersections are what solving a linear system computes, and neither of them is a perpendicular. The vector is the same on both sides.

12.4 Gram–Schmidt

Orthonormal bases are worth having; the following construction shows they always exist and produces one from any basis at all.

Theorem 12.10 (Gram–Schmidt). *Let $v_1, \dots, v_n$ be linearly independent in an inner product space. Define recursively*

$$w_k = v_k - \sum_{i<k} \langle v_k, e_i\rangle\, e_i, \qquad e_k = \frac{w_k}{\|w_k\|}.$$

Then every $w_k \neq \mathbf{0}$, the set $e_1, \dots, e_n$ is orthonormal, and for every k,

$$\operatorname{span}\{e_1, \dots, e_k\} = \operatorname{span}\{v_1, \dots, v_k\}.$$

In particular every finite-dimensional inner product space has an orthonormal basis.

Proof. Induct on k. Suppose $e_1, \dots, e_{k-1}$ are orthonormal and span the same subspace as $v_1, \dots, v_{k-1}$. Then $w_k \neq \mathbf{0}$: otherwise v_k would equal $\sum_{i<k} \langle v_k, e_i \rangle e_i$, placing it in $\operatorname{span}\{v_1, \dots, v_{k-1}\}$ and contradicting independence. For $j < k$,

$$\langle w_k, e_j \rangle = \langle v_k, e_j \rangle - \sum_{i<k} \langle v_k, e_i \rangle \langle e_i, e_j \rangle = \langle v_k, e_j \rangle - \langle v_k, e_j \rangle = 0,$$

since only the term $i = j$ survives. So e_k has norm 1 and is orthogonal to its predecessors. Finally e_k is a combination of v_k and $e_1, \dots, e_{k-1}$, and conversely v_k is a combination of $e_1, \dots, e_k$, so the two spans agree at every stage. The last sentence follows by applying the construction to any basis. □

One step of that induction is all there is to see, and Figure 12.2 shows it: subtract from v_2 its component along e_1, and what is left meets v_1 at a right angle by construction.

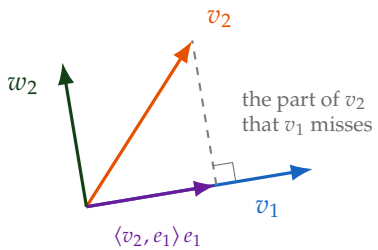


Figure 12.2. One step of Theorem 12.10. Subtracting from v_2 its component along e_1 leaves w_2, which is perpendicular to v_1 by construction: the dashed leg and the right angle are the whole of the calculation. Normalising w_2 gives e_2. In higher dimensions the same step is repeated, subtracting the components along all the directions already fixed.

Example 12.11 (Gram–Schmidt in $\mathbb{R}^3$)**.** *Orthonormalise $v_1 = (1, 1, 0)$, $v_2 = (1, 0, 1)$, $v_3 = (0, 1, 1)$, which are independent since the matrix with these columns has determinant -2.*

Step 1. $w_1 = (1, 1, 0)$, $\|w_1\| = \sqrt{2}$, *so* $e_1 = \frac{1}{\sqrt{2}}(1, 1, 0)$.

Step 2. $\langle v_2, e_1\rangle = 1/\sqrt{2}$, *so*

$$w_2 = (1,0,1) - \tfrac{1}{\sqrt{2}}\cdot\tfrac{1}{\sqrt{2}}(1,1,0) = \left(\tfrac{1}{2}, -\tfrac{1}{2}, 1\right), \qquad \|w_2\| = \sqrt{\tfrac{3}{2}},$$

giving $e_2 = \frac{1}{\sqrt{6}}(1,-1,2)$.
Step 3. $\langle v_3, e_1\rangle = 1/\sqrt{2}$ *and* $\langle v_3, e_2\rangle = 1/\sqrt{6}$, *so*

$$w_3 = (0,1,1) - \tfrac{1}{2}(1,1,0) - \tfrac{1}{6}(1,-1,2) = \left(-\tfrac{2}{3}, \tfrac{2}{3}, \tfrac{2}{3}\right), \qquad \|w_3\| = \tfrac{2}{\sqrt{3}},$$

giving $e_3 = \frac{1}{\sqrt{3}}(-1,1,1)$.
Check the three inner products: $\langle e_1, e_2\rangle = (1{-}1{+}0)/\sqrt{12} = 0$, $\langle e_1, e_3\rangle = (-1{+}1{+}0)/\sqrt{6} = 0$ *and* $\langle e_2, e_3\rangle = (-1-1+2)/\sqrt{18} = 0$; *and each norm is* 1, *since* $1+1=2$, $1+1+4=6$ *and* $1+1+1=3$. *Perform this check every time: it costs three dot products and catches every arithmetic slip.*

12.5 The QR factorisation

Gram–Schmidt expresses each v_k in terms of $e_1, \dots, e_k$ and no others. Written as matrices, that triangular pattern is a factorisation.

Theorem 12.12 (QR). *Let A be $m \times n$ with linearly independent columns. Then*

$$A = QR$$

where Q is $m \times n$ with orthonormal columns and R is $n \times n$ upper triangular with positive diagonal entries. The factorisation is unique.

Proof. Apply Theorem 12.10 to the columns $v_1, \dots, v_n$ of A and let Q have the resulting $e_1, \dots, e_n$ as its columns. Rearranging the recursion,

$$v_k = \|w_k\|\, e_k + \sum_{i<k} \langle v_k, e_i\rangle\, e_i,$$

so v_k is the combination of the columns of Q with coefficients $r_{ik} = \langle v_k, e_i\rangle$ for $i < k$, $r_{kk} = \|w_k\| > 0$, and $r_{ik} = 0$ for $i > k$. Collecting these into the columns of R gives $A = QR$ by Proposition 4.4, and R is upper triangular with positive diagonal.

For uniqueness, suppose $Q_1R_1 = Q_2R_2$ and put $S = R_2R_1^{-1}$, which is upper triangular with positive diagonal because R_1^{-1} is. Then $Q_1 = Q_2S$, and since Q_1 and Q_2 both have

orthonormal columns,

$$\mathbf{1} = Q_1^t Q_1 = S^t Q_2^t Q_2 S = S^t S.$$

(Note that $Q_2 Q_2^t$ is *not* the identity when $m > n$, which is why the argument runs in this order.) So S is an upper triangular matrix with orthonormal columns and positive diagonal. Comparing columns from the first onwards forces $S = \mathbf{1}$: column 1 is $s_{11} e_1$ of norm 1 with $s_{11} > 0$, so $s_{11} = 1$; and if columns $1, \dots, k-1$ are $e_1, \dots, e_{k-1}$, then orthogonality of column k to those kills $s_{1k}, \dots, s_{k-1,k}$, leaving $s_{kk} e_k$ of norm 1 with $s_{kk} > 0$. Hence $R_2 = R_1$ and $Q_1 = Q_2$. □

Figure 12.3 shows the shape of the factorisation. The columns of Q are the orthonormal vectors the process produces, and R records how each original column was built from the ones fixed before it, which is why it comes out triangular.

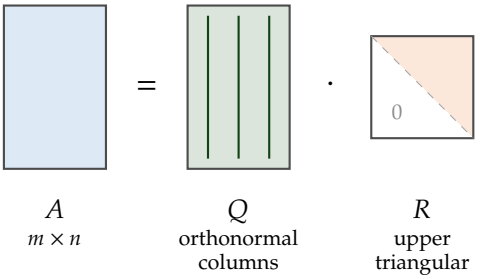


Figure 12.3. The shape of Theorem 12.12. The columns of Q are the orthonormal vectors Gram–Schmidt produces, drawn here as vertical strokes, and R records how each original column was built from the ones fixed before it. Its triangularity is not an accident of the algorithm; it is the statement that step k never uses a direction discovered later.

Example 12.13 (The QR of the previous example)**.** *With A the matrix whose columns are the v_i of Example 12.11,*

$$Q = \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{6} & -1/\sqrt{3} \\ 1/\sqrt{2} & -1/\sqrt{6} & 1/\sqrt{3} \\ 0 & 2/\sqrt{6} & 1/\sqrt{3} \end{pmatrix}, \qquad R = \begin{pmatrix} \sqrt{2} & 1/\sqrt{2} & 1/\sqrt{2} \\ 0 & \sqrt{6}/2 & 1/\sqrt{6} \\ 0 & 0 & 2/\sqrt{3} \end{pmatrix}.$$

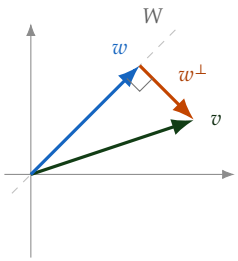
The diagonal of R holds the three norms $\|w_k\|$ and the entries above it are the coefficients $\langle v_k, e_i \rangle$ already computed. As a check, $\det R = \sqrt{2} \cdot \frac{\sqrt{6}}{2} \cdot \frac{2}{\sqrt{3}} = 2$, and $|\det A| = 2$, consistent with $|\det Q| = 1$.

Caution

The recursion of Theorem 12.10, implemented literally, is numerically poor. Rounding errors make each new e_k slightly non-orthogonal to its predecessors, and because every later step subtracts components along all the earlier ones, the error compounds; for an ill-conditioned A the computed columns of Q can lose orthogonality completely. Two standard repairs exist. *Modified* Gram–Schmidt subtracts the components one at a time, updating the working vector after each subtraction rather than using the original v_k throughout, which is the same in exact arithmetic and far better in floating point. Householder reflections abandon the recursion entirely and build Q as a product of reflections; that is what a library does.

12.6 Orthogonal and unitary matrices

Before turning to the matrices that preserve an inner product, it is worth naming what the last two sections have been doing. Figure 12.4 shows it in the plane: every vector splits, in exactly one way, into a piece lying in a subspace and a piece perpendicular to it, and Pythagoras then relates the three lengths. That splitting is the whole of Chapter 13.



$v \;=\; (3,1) \;=\; (2,2) \,+\, (1,-1)$, and $10 \;=\; 8 \,+\, 2$

Figure 12.4. Every vector splits, in exactly one way, into a piece inside a subspace and a piece perpendicular to it. Here W is the line through $(1,1)$ and $v = (3,1)$ decomposes as $(2,2) + (1,-1)$, the second summand meeting W at a right angle. Pythagoras then gives $\|v\|^2 = \|w\|^2 + \|w^\perp\|^2$, that is $10 = 8 + 2$, with no cross term precisely because the two pieces are orthogonal. Uniqueness is what makes the projection a well defined linear map rather than a choice, and it is the reason $V = W \oplus W^\perp$ for every subspace of an inner product space. Chapter 13 is this picture with W a column space, and Chapter 26 is what survives when the inner product is taken away.

Definition 12.14 (Orthogonal matrix)**.** *A square real matrix Q is* orthogonal *if $Q^tQ = \mathbf{1}$, equivalently if its columns form an orthonormal basis of $\mathbb{R}^n$. Then $Q^{-1} = Q^t$.*

Proposition 12.15 (Orthogonal maps preserve the geometry)**.** *For a real square Q the following are equivalent: Q is orthogonal; $\langle Qx, Qy\rangle = \langle x, y\rangle$ for all x, y; $\|Qx\| = \|x\|$ for all x. Moreover* $\det Q = \pm 1$.

Proof. If $Q^tQ = \mathbf{1}$ then $\langle Qx, Qy\rangle = (Qy)^tQx = y^tQ^tQx = y^tx = \langle x, y\rangle$, and taking $y = x$ preserves norms. Conversely, if norms are preserved then so are inner products, by the identity $\langle x, y\rangle = \frac{1}{2}(\|x+y\|^2 - \|x\|^2 - \|y\|^2)$; and then $\langle (Q^tQ - \mathbf{1})x, y\rangle = 0$ for all x, y, which forces $Q^tQ = \mathbf{1}$. The determinant statement is Exercise 8.26. □

Key Idea

This is why orthogonal changes of basis are the ones numerical analysts want. A general change of basis $P^{-1}AP$ can magnify errors by the condition number of P, as the Caution in Section 10.4 showed. An orthogonal change of basis Q^tAQ cannot magnify anything at all, because Q preserves every length exactly. Chapter 9 promised that the QR algorithm converges to a triangular form through changes of basis that cost nothing in accuracy; this proposition is the reason they cost nothing.

The complex case

Over $\mathbb{C}$ the definition needs one change, because $\sum_i z_i^2$ is not a sensible notion of squared length: it can be negative or complex. Replacing z_i^2 by $|z_i|^2 = z_i\bar{z}_i$ fixes it, at the cost of symmetry.

Definition 12.16 (Complex inner product, adjoint, unitary)**.** *A* complex inner product *on a complex vector space satisfies $\langle v, u\rangle = \overline{\langle u, v\rangle}$, is linear in its first argument, and has $\langle v, v\rangle > 0$ for $v \neq \mathbf{0}$. The standard one on $\mathbb{C}^n$ is $\langle x, y\rangle = \sum_i x_i\bar{y}_i$. For a complex matrix, the* adjoint *is $A^* = \overline{A}^t$, and U is* unitary *if $U^*U = \mathbf{1}$, equivalently if its columns are orthonormal for this product.*

Conjugate symmetry forces $\langle v, v\rangle$ to be real, so the positivity requirement makes sense, and it makes the form *conjugate*-linear in the second argument: $\langle u, \lambda v\rangle = \bar{\lambda}\langle u, v\rangle$. Everything in this chapter goes through with ^{t} replaced by * and orthogonal replaced by unitary, including Cauchy–Schwarz, Gram–Schmidt, QR and Proposition 12.15.

In particular the word promised in Chapter 10 is now defined, and Schur's theorem $A = QTQ^*$ can be stated precisely; we prove it next, on the way to the spectral theorem.

Insight

An inner product looks like extra structure bolted onto a vector space, and in a sense it is: a vector space has no notion of length or angle until one is chosen. What is surprising is how much follows from so little.

The axioms ask only for a symmetric, bilinear, positive definite pairing. From that alone come Cauchy and Schwarz, hence angles; the triangle inequality, hence a metric; orthogonality, hence projection; Gram and Schmidt, hence orthonormal bases; and the QR factorisation, hence a stable way to solve least squares. None of those were assumed. They are consequences of positivity, which is the only axiom with any content, and the reason is that positivity turns $\langle x - ty, x - ty \rangle \geq 0$ into a quadratic in t with no real roots, and a discriminant condition is an inequality.

That pattern, a positivity hypothesis becoming a quadratic with no roots becoming an inequality, recurs throughout this book. It is how Cauchy and Schwarz is proved here, how the arithmetic and geometric mean inequality is proved in Chapter 37, and in a matrix-valued form it is how Chapter 15 characterises positive definiteness by leading minors. If you remember one mechanism from this chapter, remember that one.

Open Problem 12.17 (Mutually unbiased bases in dimension six)**.** *Two orthonormal bases $\{u_i\}$ and $\{v_j\}$ of $\mathbb{C}^d$ are* mutually unbiased *if $|\langle u_i, v_j \rangle|^2 = 1/d$ for every i and j: measured in one basis, a vector of the other is maximally spread out.* How many pairwise mutually unbiased bases exist in $\mathbb{C}^6$?

What is known. *There can never be more than $d + 1$, an upper bound that follows from a dimension count in the real vector space of Hermitian matrices, and $d + 1$ is achieved whenever d is a prime power, by an explicit construction from finite fields. Six is the first dimension that is not a prime power. Three mutually unbiased bases are known in $\mathbb{C}^6$; a fourth has never been found, and extensive numerical searches suggest none exists [205, 62].* Nobody has proved that four do not exist, and nobody has found a fourth. *The problem has been open since the 1980s.*

Where the difficulty sits. *The upper bound of $d + 1$ is a clean linear algebra argument: the $d^2 - 1$ traceless projections coming from m mutually unbiased bases are orthogonal in the Hilbert and Schmidt inner product, forcing $m(d-1) \leq d^2 - 1$. That argument is blind to arithmetic, and it is arithmetic that appears to decide the answer. Every construction that attains the bound uses a field of order d, and when no such field exists there is nothing to substitute.*

Where a reader might start. *Build all three mutually unbiased bases in $\mathbb{C}^2$ explicitly: they are the eigenbases of the three Pauli matrices, and checking the $\left|\langle u_i, v_j\rangle\right|^2 = 1/2$ condition is twelve short computations, four for each of the three pairs of bases. Then do $\mathbb{C}^3$, where four bases exist and can be written down with cube roots of unity. Then attempt a fourth in $\mathbb{C}^6$ and watch the equations refuse to close. Chapter 51 explains why physicists care, and Chapter 41 is where the angles between these bases are given their proper name.*

Exercises

Exercise 12.18. *Which of these are inner products on the given space? For those that are not, say which axiom fails.*

(a) $\langle x, y\rangle = 2x_1y_1 + 3x_2y_2$ *on* $\mathbb{R}^2$.

(b) $\langle x, y\rangle = x_1y_1 - x_2y_2$ *on* $\mathbb{R}^2$.

(c) $\langle x, y\rangle = x_1y_2 + x_2y_1$ *on* $\mathbb{R}^2$.

(d) $\langle A, B\rangle = \operatorname{trace}(AB)$ *on* $M_2(\mathbb{R})$.

Exercise 12.19. *Apply Gram–Schmidt to* $(1,1,1)$, $(1,1,0)$, $(1,0,0)$ *in* $\mathbb{R}^3$, *and verify that the result is orthonormal.*

Exercise 12.20. *In* $C[-1,1]$ *with* $\langle f, g\rangle = \int_{-1}^{1} f\,g$, *apply Gram–Schmidt to* $1, t, t^2$. The answers, up to scaling, are the first three Legendre polynomials.

Exercise 12.21. *Find the QR factorisation of* $A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}$, *and check that* $Q^{t}Q = \mathbf{1}_2$ *and* $QR = A$.

Exercise 12.22. *Let* (e_1, e_2, e_3) *be an orthonormal basis of* $\mathbb{R}^3$ *and suppose* $\langle v, e_1\rangle = 2$, $\langle v, e_2\rangle = -1$ *and* $\|v\| = 3$. *Find* $|\langle v, e_3\rangle|$, *and explain why its sign is not determined.*

Exercise 12.23. *For a subset S of an inner product space, define* $S^{\perp} = \{v : \langle v, s\rangle = 0 \text{ for all } s \in S\}$. *Show that $S^{\perp}$ is a subspace even when S is not, that* $S^{\perp} = (\operatorname{span} S)^{\perp}$, *and that* $S \cap S^{\perp} \subseteq \{\mathbf{0}\}$.

Exercise 12.24. *Show that the product of two orthogonal matrices is orthogonal and that the inverse of an orthogonal matrix is orthogonal. Then show that every 2×2 orthogonal matrix is either a rotation or a reflection, according to the sign of its determinant.*

Exercise 12.25. *Use Theorem 12.3 in $\mathbb{R}^n$ with the vector $(1, 1, \dots, 1)$ to prove that $\left(\sum_i x_i\right)^2 \le n \sum_i x_i^2$ for all real x_i, and say when equality holds.*

Exercise 12.26. *Verify that $\langle A, B\rangle = \operatorname{trace}(A^t B)$ satisfies all three axioms on $M_{m\times n}(\mathbb{R})$. What is $\|A\|$ in this inner product, written in terms of the entries, and why is $\operatorname{trace}(AB)$ from Exercise 12.18(d) not a substitute?*

References and further reading

The misattribution of Gram–Schmidt is documented in the historical literature and is a useful reminder that names in mathematics record who was read, not who was first. Axler [37] is excellent on this material and, characteristically, insists on separating what needs an inner product from what does not, which is exactly the distinction this chapter turns on. Treil [610] is good on the complex case and on why conjugate symmetry is forced rather than chosen. Hefferon [307] and Austin [35] both give more computational practice with Gram–Schmidt than we have room for. Banica [51], our baseline, sets up inner products in essentially this way and proceeds directly to the spectral theorems, which is where the next chapters go. For the numerical side, and in particular for the difference between classical and modified Gram–Schmidt and for why Householder reflections displaced both, Trefethen and Bau [607] devote several lectures to QR alone; their demonstration of how badly classical Gram–Schmidt can lose orthogonality is worth running once for yourself.